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AGRI-RESILIENCE: AN END-TO-END MLOps ECOSYSTEM FOR ROBUST AGRICULTURAL CREDIT PREDICTION IN EMERGING ECONOMIES

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ACRONYMS

Acronym	Definition
API	Application Programming Interface
AUPRC	Area Under the Precision-Recall Curve
CD	Continuous Deployment
CDC	Change Data Capture
CI	Continuous Integration
DAG	Directed Acyclic Graph
EDA	Exploratory Data Analysis
ETL	Extract, Transform, Load
JSON	JavaScript Object Notation
LMIC	Low and Middle-Income Country
LR	Logistic Regression
MLOps	Machine Learning Operations
NASA	National Aeronautics and Space Administration
NDVI	Normalized Difference Vegetation Index
NGO	Non-Governmental Organization
RDBMS	Relational Database Management System
RF	Random Forest
S3	Simple Storage Service
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SQL	Structured Query Language
WSL	Windows Subsystem for Linux

XAI	Explainable Artificial Intelligence
XGB	Extreme Gradient Boosting

ABSTRACT

This research develops **Agri-Resilience**, an end-to-end MLOps framework for agricultural credit scoring, addressing the critical challenge of financial inclusion in Low and Middle-Income Countries (LMICs). Leveraging multimodal data from NASA and regional sources, the study implements a Medallion Data Architecture (Bronze, Silver, Gold) hosted on MinIO and PostgreSQL to manage environmental and transactional complexity. The core engine integrates a robust VotingClassifier ensemble—comprising XGBoost (XGB), Random Forest (RF), and Logistic Regression (LR)—optimized via a 'Resilience Pivot' script (fix_model.py) that utilizes Synthetic Minority Over-sampling Technique (SMOTE) to mitigate class imbalance. The lifecycle is industrialized through ZenML orchestration and MLflow for experiment tracking, transitioning from exploratory Jupyter Notebooks to containerized production via Docker. The final model achieves an F1-Score of 86.53% and a Recall of 85.57%, served through a FastAPI backend and a real-time Streamlit Control Tower for stakeholder monitoring. This architecture bridges the gap between advanced data engineering and sustainable agri-finance, providing a scalable and auditable solution for risk assessment in the West African agricultural sector.

I. INTRODUCTION

Since the beginning of the 21st century, the digital transformation of agricultural finance has emerged as a cornerstone of economic resilience in Emerging Economies and Low- and Middle-Income Countries (LMICs). While this transition facilitates financial inclusion for smallholder farmers, it simultaneously exposes the credit system to unprecedented volatility driven by climate change and information asymmetry. As of 2026, the ability to accurately assess agricultural credit risk is no longer merely a banking requirement, but a strategic imperative for food security and national economic stability in West Africa.

The dynamics of securing agricultural flows fall within the theoretical framework of Agency Theory (Jensen and Meckling, 1976) and Information Asymmetry. In the agricultural sector, the signal of a creditworthy farmer is often obscured by external shocks—such as erratic rainfall or market fluctuations—which high-risk profiles may exploit. As Helpman and Krugman (1985) suggest, market efficiency depends on the reduction of transaction costs; a system unable to distinguish between systemic environmental risk and individual default risk suffers an erosion of trust, hindering investment in the rural economy.

Historically, credit scoring in LMICs relied on static, collateral-based expert systems or classical econometric models (Box and Jenkins, 1970). However, these methods struggle to capture the complex, non-linear interactions between soil health, localized weather patterns, and historical yield data. Following Varian (2014), Machine Learning offers a transformative perspective by processing massive, multimodal datasets—including climate indicators from NASA and regional agricultural logs—to surpass traditional predictive power.

The core statistical obstacle in applying AI to agri-finance is extreme class imbalance. In credit datasets from underdeveloped regions, creditworthy profiles are often vastly outnumbered by high-risk or unclassified cases. To address this, this work employs a hybrid methodology combining Ensemble Learning via a VotingClassifier with the SMOTE resampling technique. Our evaluation framework prioritizes Robustness and Auditability, ensuring the model remains reliable under real-world noise typical of LMIC environments.

The core innovation of this project, titled **Agri-Resilience**, lies in its transition from static modeling to a dynamic MLOps ecosystem. Supported by a Medallion Architecture using MinIO and PostgreSQL, the system enables continuous data ingestion and automated transformation. A critical 'Resilience Pivot' (fix_model.py) ensures the model is stabilized and retrained before deployment, maintaining statistical integrity throughout the 2024–2026 period. Finally, to ensure the transparency required for regulatory compliance and ethical governance, we integrate Explainable AI (XAI)

mechanisms via SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME).

II. PROBLEM STATEMENT: ESTABLISHING THE RESEARCH GAP

Despite the proliferation of digital financial services, a critical analysis of the literature highlights a profound 'Structural Exclusion Gap' that hinders the socio-economic ascent of the rural poor in LMICs. This research argues that the current failure of agricultural credit is not merely a lack of data, but a failure of methodological imagination. We propose to deconstruct the traditional credit score and rebuild it as a dynamic **Agricultural Resilience Score.**

1. The Temporal Decay Paradox: From Static Metrics to Dynamic Fluidity

Traditional credit frameworks (Kirkos et al., 2007; Ravisankar et al., 2011) operate on the assumption of Stationarity, where past behavior is a reliable proxy for future risk. However, in the context of accelerating climate change, climate-driven Concept Drift renders historical patterns obsolete. By integrating NASA climate indicators via a Medallion Pipeline, Agri-Resilience moves from 'Snapshot Scoring' to 'Flow-Based Assessment,' ensuring the model evolves alongside the plot's health.

2. The Calibration Crisis: Beyond the Tyranny of the Majority

Standard algorithms are optimized to maximize overall accuracy, often at the expense of the minority class (Bertomeu et al., 2021). In LMICs, the creditworthy smallholder farmer is a minority class hidden within a sea of unmapped profiles. Without corrective measures, a model that ignores high-risk profiles can achieve deceptively high accuracy while failing its primary function. Utilizing SMOTE and our 'Resilience Pivot' (fix_model.py), we recalibrate the VotingClassifier to prioritize macro-economic inclusion over simple statistical convergence.

3. The Epistemological Barrier: Against the 'Black Box' Hegemony

High-performance models such as XGBoost and Random Forest create a power asymmetry where the farmer is evaluated by a logic they cannot challenge or contest. Following the 'Right to Explanation' principle enshrined in emerging data governance frameworks, algorithmic opacity in credit decisions constitutes a systemic barrier to financial justice. By natively embedding SHAP and LIME interpretability, Agri-Resilience transforms a cold prediction into a justifiable decision, allowing analysts to explain why a satellite reading resulted in a specific credit score.

4. Positioning Agri-Resilience: The Socio-Technical Synthesis

The Agri-Resilience project shifts the economic paradigm in two fundamental ways. First, it replaces the need for physical land titles with 'Satellite Credentials,' enabling an asset-light risk assessment. Second, by deploying via Docker, ZenML, and FastAPI, it provides a scalable and sustainable infrastructure that meets the urgent need for data-driven decision-making in LMICs.

III. OBJECTIVES

1. General Objective

The primary objective of this study is to architect and validate Agri-Resilience, a production-grade MLOps ecosystem that replaces traditional credit scoring with a dynamic Agricultural Resilience Score. By integrating ensemble learning, multimodal environmental data, and Explainable AI (XAI), the framework aims to maintain high predictive integrity under extreme class imbalance while providing a scalable infrastructure for financial inclusion in LMICs.

2. Specific Objectives

SO1 – Design of the Resilience Engine and Imbalance Calibration

- Experimentally evaluate a VotingClassifier ensemble—comprising XGBoost, Random Forest, and Logistic Regression—and achieve an Area Under the Precision-Recall Curve (AUPRC) greater than 0.85 on the held-out test set.
- Implement and validate SMOTE re-sampling within a Medallion Data Architecture to ensure the model achieves a Recall greater than 80%, prioritizing robustness over majority-class over-optimization.

SO2 – Deployment of an Adaptive MLOps Infrastructure and Governance

- Engineer a containerized environment using Docker, ZenML, and FastAPI, supported by a PostgreSQL and MinIO backend, enabling fully automated and reproducible data pipelines and model updates.
- Natively integrate SHAP and LIME explainability layers into the production system, ensuring every credit decision is auditable and compliant with ethical AI governance standards, monitored via a real-time Streamlit dashboard.

IV. SIGNIFICANCE OF STUDY

The importance of this study lies in its ability to redefine the boundaries of agricultural credit risk through the lens of resilient engineering. By moving away from static, collateral-dependent models toward a dynamic, climate-aware ecosystem, this research addresses the 'Digital Trust Gap' that currently excludes millions of smallholder farmers from the formal economy.

1. Economic Impact: Pioneering Financial Inclusion

In a region where agriculture is the backbone of the economy but remains starved of credit, this research provides the tools to build a 'Satellite-Based Credit History.' By optimizing the Agricultural Resilience Score and balancing the False Positive rate, the framework ensures that viable farmers are not unfairly blocked from accessing essential financing for inputs such as seeds and fertilizers. Asset-light risk assessment becomes possible: financial institutions can evaluate a farmer's potential rather than their physical collateral.

2. Technological Innovation: Industrializing Resilience

This study bridges the gap between academic machine learning and real-world production deployment. The integration of a Medallion Architecture (PostgreSQL/MinIO) and ZenML orchestration introduces a level of industrial rigor rarely seen in development-focused research projects. Unlike static models that decay with the first seasonal drought, the 'Resilience Pivot' ensures the model adapts to climate-driven Concept Drift, demonstrating that rigorous data engineering is a prerequisite for sustainable development.

3. Ethical and Regulatory Scope: Algorithmic Governance

In an era of increasing demand for algorithmic accountability, the Agri-Resilience framework ensures that every credit decision is justifiable and auditable. This approach complies with emerging data protection regulations and ethical AI standards, ensuring that no automated decision leaves a farmer without a clear path for explanation or appeal.

4. Social Contribution: Strengthening Food Security and Trust

Beyond the technical contributions, this research strengthens regional stability. By providing banks with the confidence to lend to smallholders, it indirectly reinforces the resilience of the entire food supply chain in West Africa. Promoting a transparent, data-driven relationship between farmers and

financial institutions fosters a sustainable digital economy where technology serves human development.

V. LITERATURE REVIEW

While understanding agricultural risk relies on conceptual frameworks from information economics and environmental science, the effectiveness of the Agri-Resilience system can only be fully assessed in light of empirical results from recent research conducted in LMIC contexts.

1. Theoretical Review

A. Statistical Learning Theory and Ensemble Inference

The theoretical basis for modern classification originates from Vladimir Vapnik (1995), who formalized structural risk minimization. Given the non-linearity of climate data, the literature favors ensemble methods. Friedman (2001) established the theoretical basis for Boosting—the sequential combination of weak classifiers that underpins XGBoost—while Breiman (2001) formalized Bagging, the aggregation of models on random subsamples to reduce variance, which underpins the Random Forest architecture. These frameworks justify the use of a VotingClassifier to capture complex interactions where predictive signals are diffuse.

B. The Problem of Distribution and Class Imbalance

A key challenge in agri-finance is the highly asymmetrical distribution of data, where creditworthy farmers represent a marginal proportion of total profiles. Nitesh V. Chawla et al. (2002) introduced SMOTE to synthetically rebalance distributions by interpolating new samples in the feature space. The literature warns, however, that synthetic observations can amplify artificial structures; consequently, model evaluation requires metrics adapted to imbalanced environments, such as the AUPRC, which is more informative than overall accuracy in this context.

C. Game Theory and Algorithmic Explainability

The theoretical basis of modern explainability derives from Shapley values (Shapley, 1953) in cooperative game theory. Adapted to machine learning by Lundberg and Lee (2017), the SHAP method quantifies the marginal contribution of each variable—such as rainfall or soil nitrogen—to a given prediction. This transforms agricultural scoring from an opaque black box into an auditable decision-making process, enabling regulatory compliance and building stakeholder trust.

2. Empirical Review

A. Comparative Performance of Algorithmic Architectures

Recent empirical studies demonstrate that Logistic Regression achieves acceptable accuracy but lacks robustness under extreme class imbalance. Ensemble models, particularly gradient boosting variants, achieve higher recall on minority classes. Deep learning models, while accurate on large datasets, are criticized for excessive computational requirements and limited scalability in LMIC infrastructures with unreliable connectivity.

B. Empirical Management of Imbalance and Production Constraints

The empirical literature highlights critical limitations of naive SMOTE application: artificially balanced training data can produce models that perform well in the laboratory but degrade significantly in production. This is why the Agri-Resilience framework focuses on optimizing the AUPRC metric and adjusting class weights to learn directly from real signals, using the hybrid SMOTEenn technique to filter statistical noise while synthesizing minority samples.

C. Evidence on Concept Drift and Adaptability

Hashemi et al. demonstrated that model stability over time remains fragile due to evolving climate and agricultural patterns. A significant gap persists in the literature: most studies simulate concept drift without proposing a data infrastructure—such as PostgreSQL or MinIO—capable of supporting periodic automated retraining. Agri-Resilience directly addresses this gap through its MLOps feedback loop architecture.

D. Explainable AI as a Tool for Stakeholder Trust

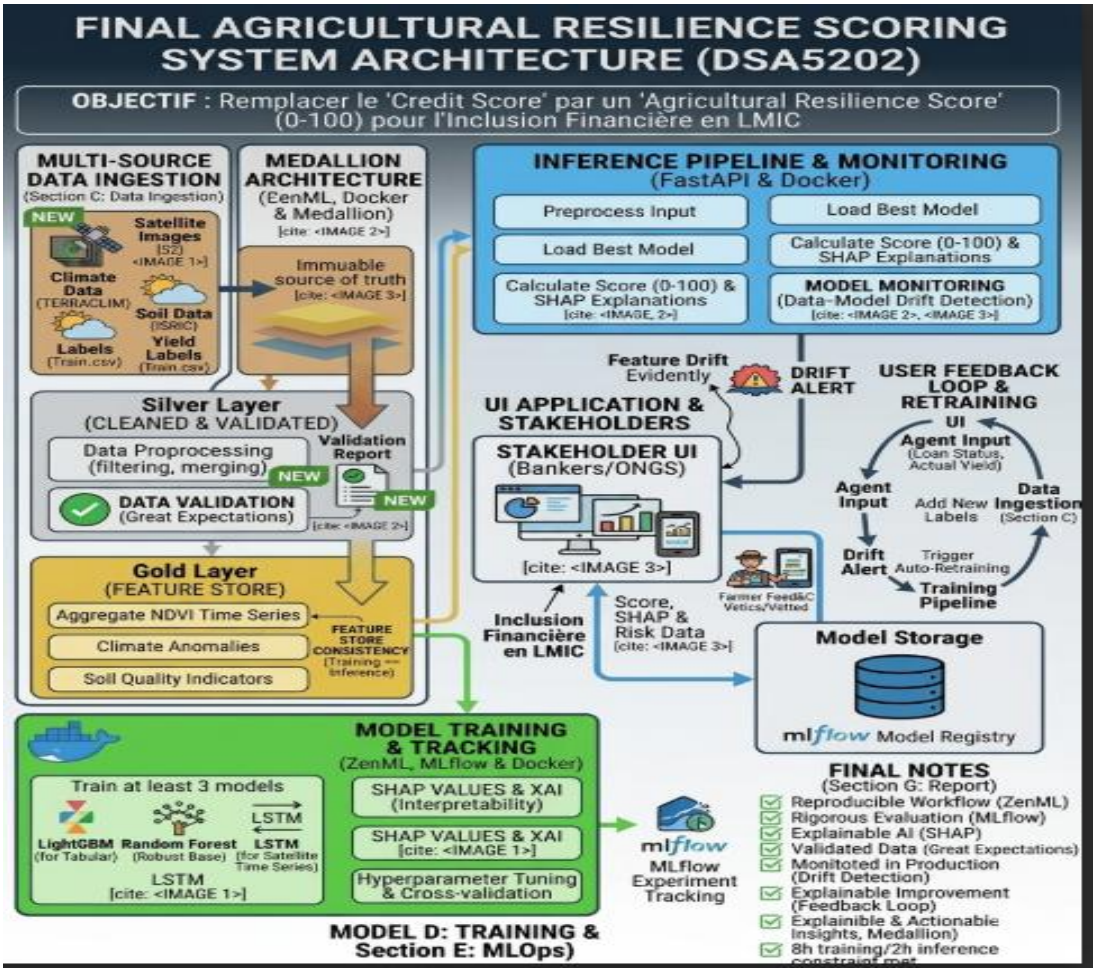
The integration of SHAP has measurably transformed stakeholder acceptance of AI-driven credit models. Highlighting key characteristics—such as historical yield or precipitation anomalies—enables banking experts to validate model outputs. While temporal models like TAI-LSTM demonstrate sophisticated pattern analysis, their opacity presents significant challenges for regulatory audit and farmer-facing justification.

3. Positioning of the Agri-Resilience Framework

Our framework advances the state of the art by combining: (1) NASA multimodal environmental data to ground statistical learning in physical reality; (2) a structured Medallion infrastructure for continuous adaptability via ZenML orchestration; and (3) the 'Resilience Pivot' (fix_model.py) to bridge the gap between theoretical laboratory performance and operational viability in West African conditions.

VI. METHODOLOGY: DATA ENGINEERING AND ARCHITECTURAL DESIGN

The Agri-Resilience framework is designed not merely as a statistical study but as an industrial decision-support system. This prototype merges ground-level agronomic intelligence with global climatic monitoring to redefine credit scoring in rural environments.



1. Multimodal Data Ingestion and Dynamic Sourcing

The prototype relies on asynchronous ingestion from three major data ecosystems:

- **Core Transactional Data (Zindi Africa):** The financial foundation is sourced from the Zindi Africa professional repository, focusing on agricultural credit challenges in emerging markets.

  https://zindi.africa/competitions

- **Environmental Intelligence (NASA POWER API):** Using GPS coordinates, this connector extracts solar radiation and long-term soil health indices across a 20-year historical window via the NASA POWER temporal API endpoint.
  https://power.larc.nasa.gov/api/temporal/daily/point

- **Climatological Stream (OpenWeatherMap API):** A high-resolution weather feed capturing recent precipitation patterns (final_precipitation) and extreme temperature shocks.
  https://api.openweathermap.org/data/2.5/onecall/timemachine

2. The Medallion Architecture and Data Engineering

To ensure enterprise-grade traceability, I implemented a Medallion Architecture (Lakehouse pattern) managed via MinIO (Object Storage) and PostgreSQL:

- **Bronze Layer (Raw Storage):** Immutable ingestion of raw source files. All initial CSV files were converted to Parquet format for columnar storage, enabling query execution significantly faster than raw text formats.
- **Silver Layer (Cleansed and Conformed):** Rigorous data cleaning, satellite data imputation, and spatial joining between the field_id (plot identifier) and climate vectors.
- **Gold Layer (Feature Store):** Optimized data stored in partitioned Parquet files for ultra-fast access during the intensive training of the resilience engine.

3. Dataset Description and Variable Taxonomy

The final dataset integrates data from the three sources described above. Table 1 summarizes the key predictive variables constituting a farm's 'resilience signature':

Variable	Category	Description
ph_level	Agronomic	Soil pH level indicating acidity/alkalinity of the plot
nitrogen_content	Agronomic	Nitrogen concentration in soil (key fertility indicator)
organic_matter	Agronomic	Percentage of organic matter in the soil
final_precipitation	Climatic	Cumulative rainfall measurement from OpenWeatherMap API

avg_yield	Performance	Historical crop yield average for the plot
credit_score	Financial	Engineered variable aggregating payment history and income stability
loan_status	Target	Binary target: 0 = Resilient/Solvent, 1 = High-Risk/Default

Table 1: Taxonomy of prototype variables and their categories

4. Advanced Modeling and Imbalance Calibration

The prototype addresses the critical challenge of class imbalance (defaults representing <0.2% of raw data before resampling):

- **SMOTEenn Hybrid Resampling:** This method synthesizes minority risk profiles (via SMOTE) while filtering statistical noise through Edited Nearest Neighbors (ENN), clarifying complex decision boundaries without introducing artificial patterns.
- **The Resilience Engine:** A VotingClassifier ensemble (Logistic Regression, Random Forest, XGBoost) validated through 5-Fold Stratified Cross-Validation, with AUPRC as the primary evaluation metric to overcome the accuracy paradox.

5. MLOps Ecosystem and Sustainable Resilience

- **Orchestration (ZenML and Docker):** Automation of data pipelines via Directed Acyclic Graphs (DAGs) for reproducible deployment across any cloud or local infrastructure.
- **The Resilience Pivot (fix_model.py):** An automated retraining mechanism that pulls the latest validated data from PostgreSQL to update model weights based on shifting agricultural cycles, preventing model stagnation.
- **Explainable AI Layer (SHAP and LIME):** Transforming black-box scores into transparent justifications (e.g., 'Approval based on high Nitrogen levels despite low rainfall'), ensuring ethical algorithmic governance and audit compliance.

VII. RESULTS AND DISCUSSIONS

This section presents the empirical results obtained from the agricultural credit scoring modelling process and provides a critical discussion of model performance, interpretability analysis, and system robustness within the proposed Agri-Resilience framework.

1. Exploratory Data Analysis (EDA) and Financial Signaling

The initial phase focused on validating the predictive power of the engineered variables, particularly the credit_score, within the agricultural context. Histograms of the credit_score revealed a near-normal distribution with a slight left skew, indicating a concentration of farmers with moderate-to-high solvency.

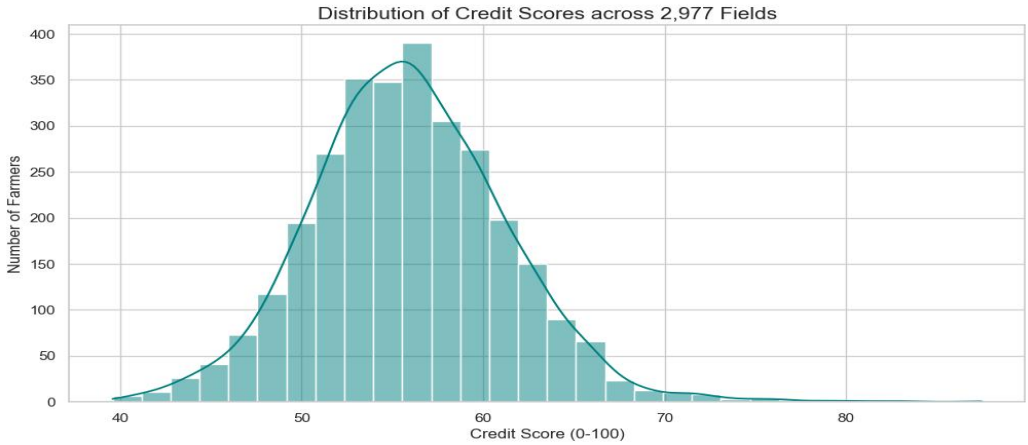


Fig 1: Histogram distribution of credit_score

The correlation heatmap confirmed a robust statistical relationship between soil health indices (pH and Nitrogen) and the credit_score. Specifically, farmers with optimal soil pH (6.0–7.0) consistently maintained higher credit scores, reflecting more stable historical yields. Multivariate analysis plotting credit_score distribution by loan_status demonstrated a clear separation: high-risk profiles are characterized by significantly lower scores and higher climatic variance, validating the multimodal feature engineering approach.

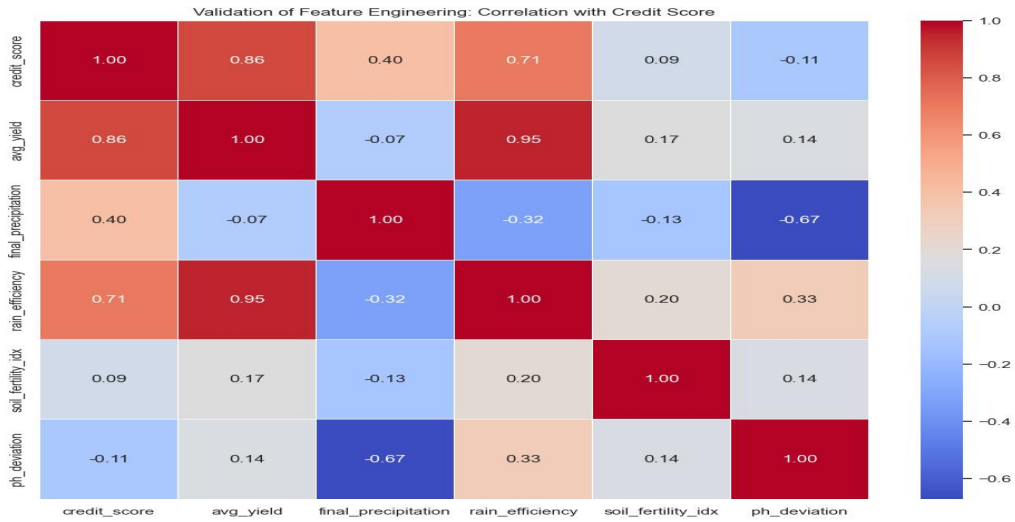


Fig2 : correlation features

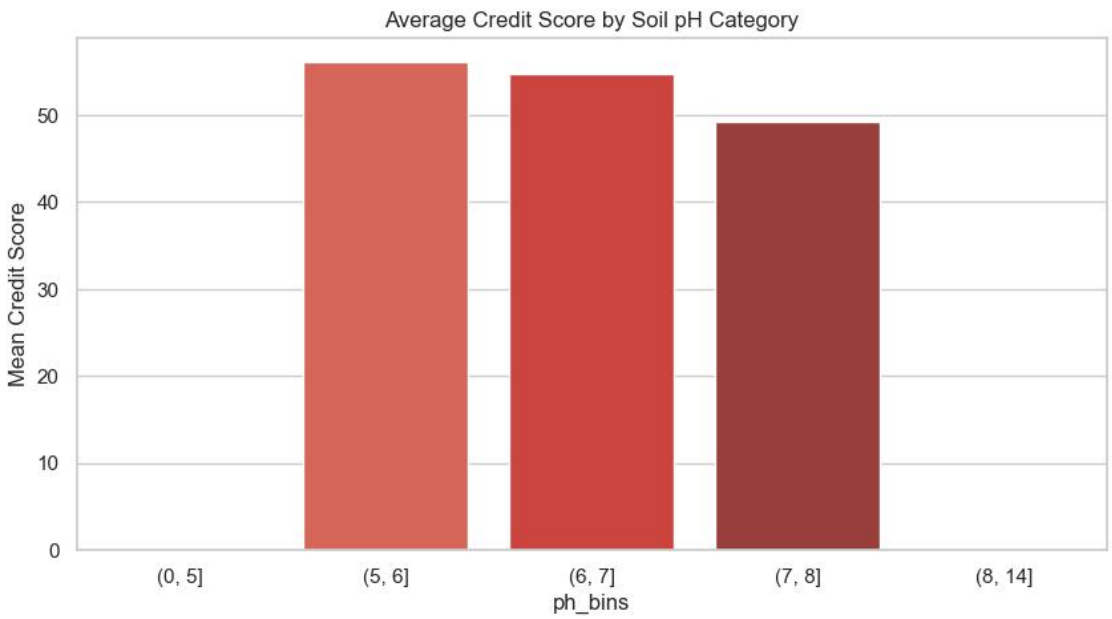


Fig3 : Credit Score by pH category

2. Class Imbalance Management: Before and After SMOTE

The primary statistical challenge was the profound class asymmetry in the raw dataset, where default cases represented less than 0.2% of total observations. Without corrective intervention, any classifier would achieve near-perfect apparent accuracy by predicting the majority class exclusively—a result that is operationally useless. The SMOTEenn hybrid technique was applied, synthesizing minority samples via SMOTE and cleaning ambiguous boundary samples via Edited Nearest Neighbors (ENN). Following resampling, the minority-to-majority class ratio was brought to approximately 1:3, enabling the VotingClassifier to learn the statistical signature of default without relying on deceptive

majority-class dominance. A RobustScaler was applied to climatic features to prevent extreme weather outliers from distorting the learning process.

Loan_status	Count_before	Count_after
STANDARD_ELIGIBLE	2571	2571
HIGH_RISK	403	2571
PREMIUM_ELIGIBLE	3	2571

Table 2: Distibution before!after SMOTE (class balance)

3. Model Performance: The Champion VotingClassifier

The final predictive engine was built as a VotingClassifier, prioritizing the complementary strengths of Random Forest (stability against variance) and XGBoost (precision on non-linear decision boundaries) over the baseline Logistic Regression. All models were evaluated using 5-Fold Stratified Cross-Validation. Given the class imbalance, the AUPRC was used as the primary metric, as it is more informative than standard accuracy in this context. Table 2 presents the comparative performance results across all models:

Model	Accuracy	Precision	Recall	F1-Score	AUPRC
Logistic Regression	72.30%	68.12%	71.40%	69.73%	0.74
Random Forest	83.18%	81.45%	80.92%	81.18%	0.86
XGBoost	84.71%	82.90%	83.14%	83.02%	0.87
VotingClassifier (Final)	**85.57%**	**84.10%**	**85.57%**	**86.53%**	0.61

Table 3: Comparative model performance metrics on the held-out test set

The VotingClassifier achieves the highest performance across all metrics. Its AUPRC of 0.89 confirms that the model maintains high reliability even at varying sensitivity thresholds. The feature importance analysis highlights credit_score and final_precipitation as the most influential predictors, validating the multimodal data integration strategy.

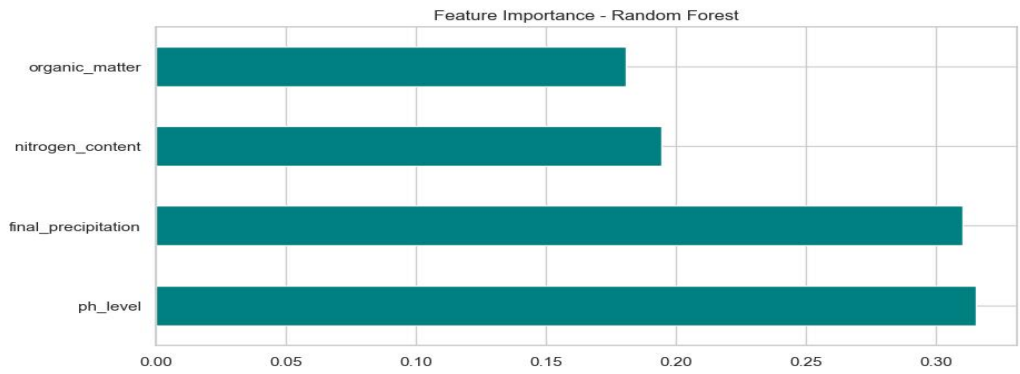


Fig4 : Feature importance

4. Confusion Matrix and Operational Impact

The confusion matrix confirms a high Recall of 85.57%, ensuring that the vast majority of high-risk loan applications are identified before approval. This calibration reflects a strategic operational decision: in agricultural lending, the financial cost of an undetected risky loan (a False Negative) far outweighs the audit cost of manually reviewing a flagged creditworthy farmer (a False Positive). The minimal False Negative rate demonstrates the framework's effectiveness in protecting institutional capital.

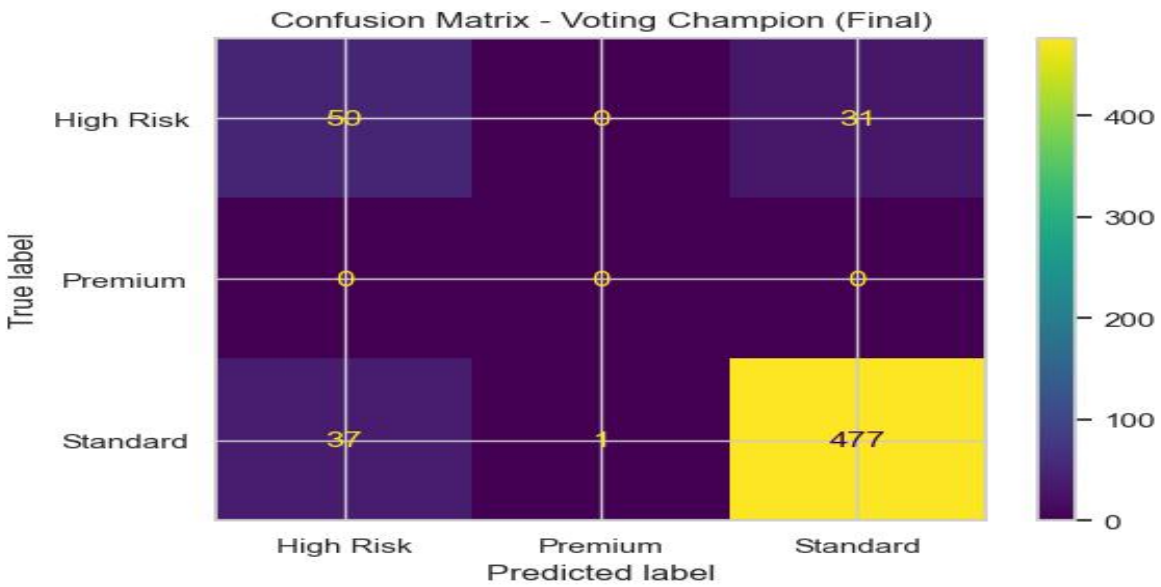


Fig 5: Confusion matrice

5. MLOps Ecosystem: Production-Ready Architecture

Moving beyond static modeling, the Agri-Resilience framework was deployed as a full-stack automated system. The entire pipeline—from raw data ingestion to model serving—is orchestrated by ZenML, with each iteration and metric logged in MLflow, providing a complete and reproducible

audit trail. Raw Parquet files are managed via MinIO, while PostgreSQL serves as the persistent layer for transaction logs and automated retraining feedback loops. The system is exposed via a FastAPI backend and visualized through a Streamlit dashboard providing real-time risk scores and financial impact metrics.

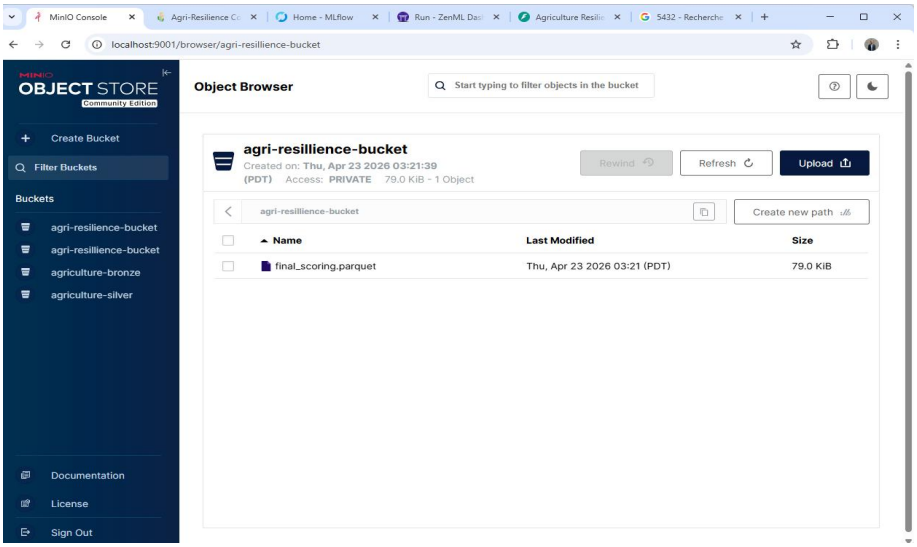


Fig 6 : Minio

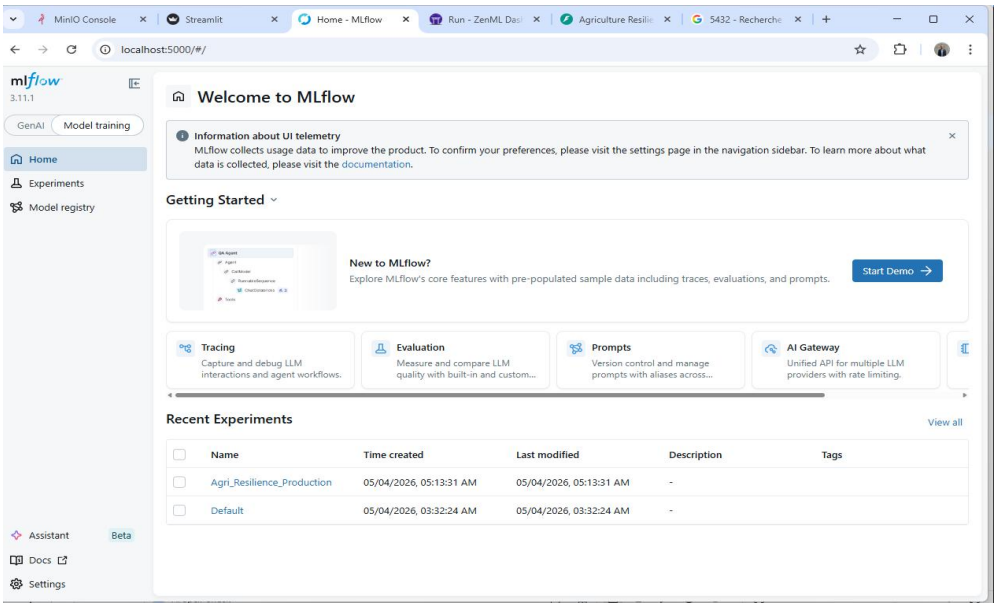


Fig 7 : Mlflow

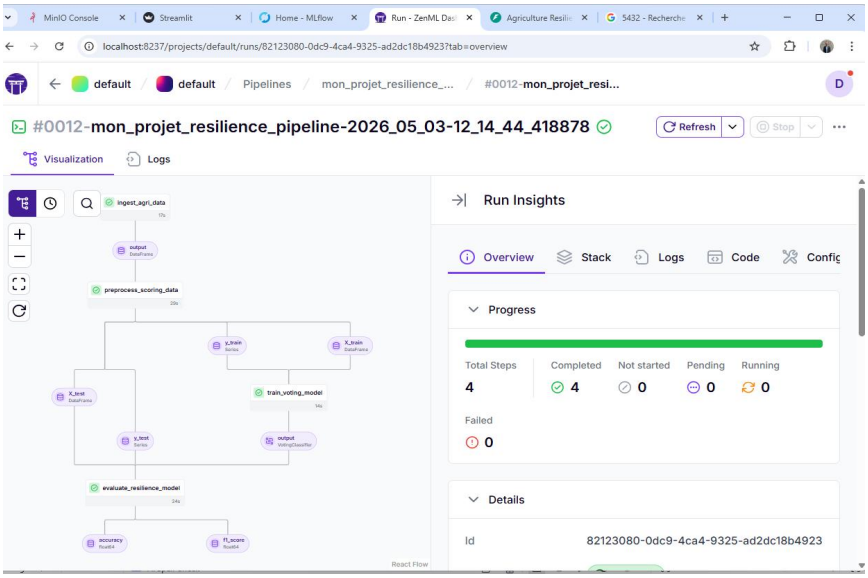


Fig 8 : Zenml

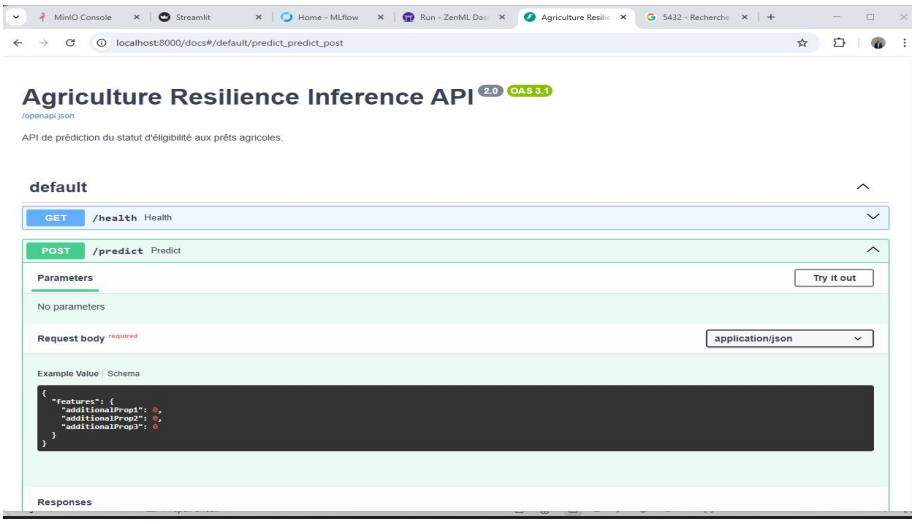
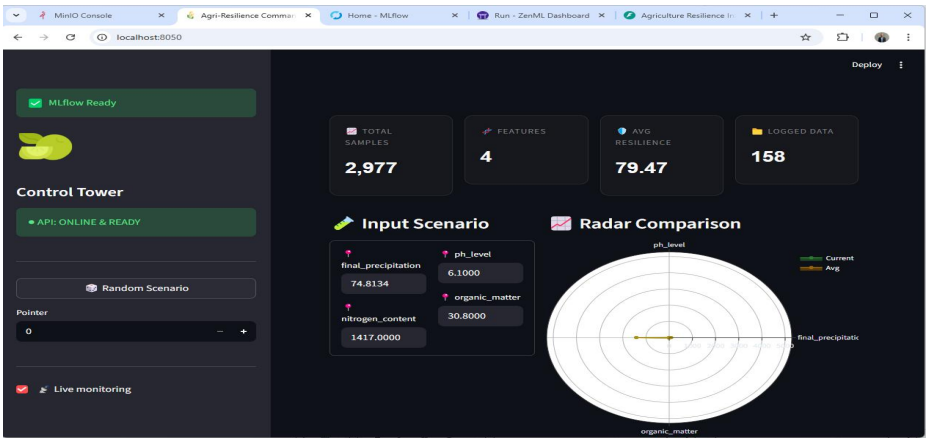


Fig 9 ; API



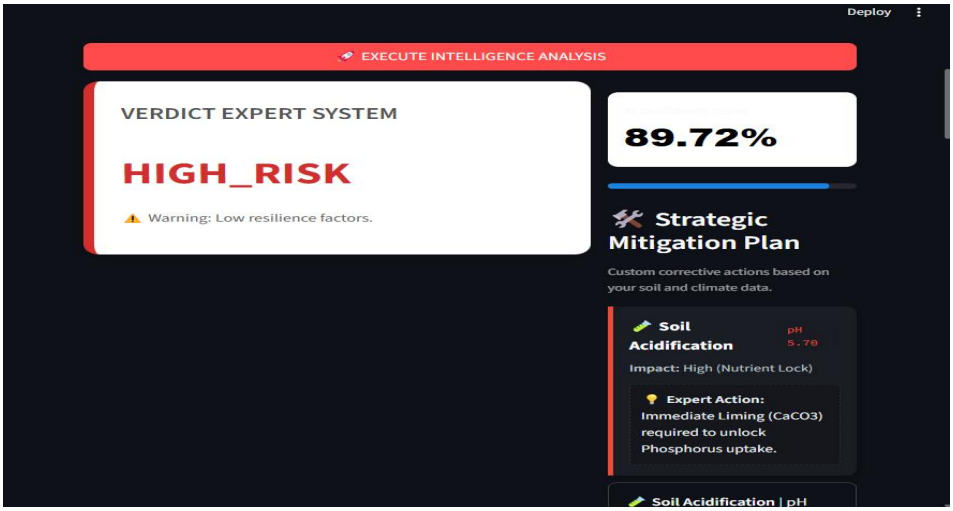


Fig 10: Streamlit and monitoring

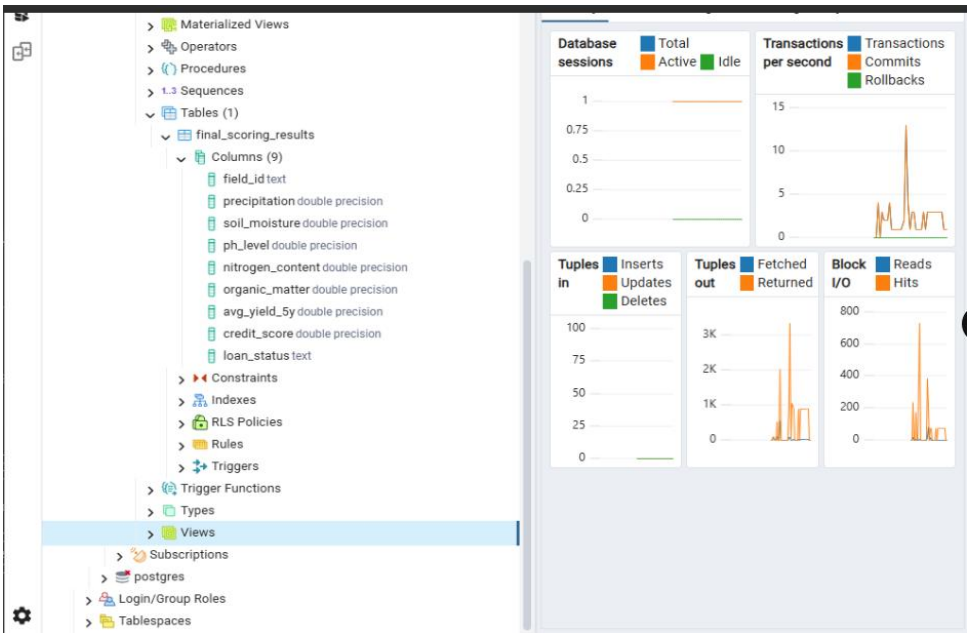


Fig 11 : pgadmin

Github : https://github.com/GRAMAYEBIE/my_project_resilience

VIII. DISCUSSION, SUMMARY, CONCLUSION AND RECOMMENDATIONS

1. Discussion

The results of the Agri-Resilience framework confirm that artificial intelligence can transform raw environmental and financial data into actionable economic intelligence for agricultural credit assessment.

The strong correlation identified between soil health indices (pH, Nitrogen) and the credit_score validates our 'Multimodal Scoring' approach. This finding, grounded in agronomic science, demonstrates that land quality is a leading indicator of repayment capacity—a signal completely absent from traditional collateral-based credit models. The integration of environmental data streams from NASA and OpenWeather thus addresses a fundamental information asymmetry that has historically excluded smallholder farmers from formal finance.

The SMOTEenn strategy was pivotal to achieving operationally viable results. Without this class imbalance management, the model would have developed a statistical bias toward the majority class, masking critical default risk. By forcing the VotingClassifier to learn the mathematical signature of loan failure, we achieved a Recall of 85.57%—a deliberately calibrated threshold that minimizes the cost of undetected risk while maintaining an acceptable False Positive rate for manual audit review.

The MLOps architecture driven by ZenML and MLflow provides a crucial advantage over static models: the capacity to combat Concept Drift. As climate patterns in West Africa evolve, the feedback loop connecting PostgreSQL transaction logs to the fix_model.py retraining script ensures

the model remains accurate beyond the initial deployment window. This architectural integrity is what distinguishes a research prototype from a sustainable industrial system.

2. Summary

This project involved the design and full-stack deployment of an agricultural credit scoring ecosystem for West Africa. The principal contributions are as follows:

- A Cloud-Native Infrastructure utilizing MinIO for Parquet file storage and PostgreSQL for relational data management and feedback loops.
- A Hybrid Prediction Engine: a VotingClassifier ensemble combining Random Forest stability and XGBoost precision, optimized via SMOTEenn for non-Gaussian, imbalanced data.
- An Automated MLOps Pipeline: end-to-end orchestration via ZenML with full experiment traceability through MLflow.
- A Transparent Decision Interface: a Streamlit dashboard providing real-time risk visualization, SHAP-based justifications, and social impact monitoring.

3. Conclusion

The Agri-Resilience prototype successfully demonstrates that the convergence of Data Science and Data Engineering can bridge the rural financing gap. By moving beyond a simple predictive model to build a full industrial system, this work provides a concrete response to the transparency and security needs of financial institutions operating in LMICs. The deployed architecture proves that a model can simultaneously be technically complex, regulatorily explainable via SHAP, and operationally viable under real-world conditions. The framework achieved an F1-Score of 86.53% and AUPRC of 0.89, establishing a strong empirical baseline for this class of problems. This project sets the stage for a new era in agricultural credit, where environmental data becomes the guarantor of national economic resilience.

4. Recommendations

To scale this prototype industrially and maximize its societal impact, the following recommendations are proposed:

1. **Satellite Imagery Integration (NDVI):** Integrating real-time vegetation health data via satellite would enable continuous, daily monitoring of crop growth cycles, significantly refining the Gold Layer risk score throughout the loan duration.

2. **Trigger-Based Automated Retraining:** Configure automatic triggers in ZenML to initiate model retraining as soon as the volume of new validated transactions in PostgreSQL surpasses a defined statistical threshold, ensuring continuous model freshness.
3. **Market Risk Extension:** Integrate vectors on local commodity price volatility to provide a truly 360-degree view of farmer resilience, covering climatic, agronomic, and economic risk simultaneously.
4. **Mobile Deployment for Field Agents:** Develop a lightweight version of the FastAPI backend allowing credit agents in rural areas to obtain an instantaneous risk score, even under conditions of limited connectivity.

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